Waseem Raza, PhD (Electrical and Computer Engineering)

Data Science & Machine Learning Scholar | Computer Science Educator

Reliable ML | Data-Efficient Learning | Time-Series Analytics | Applied Research & Student Mentoring

✉ waseem.placements@gmail.com | waseemraza844@gmail.com

New Jersey, United States ☎ +1 732-803-9703

LinkedIn | GitHub | Google Scholar | Coursera Courses | Portfolio

Academic Profile

PhD in Electrical and Computer Engineering (University of Oklahoma, 2025), specializing in AI/ML for reliable wireless systems, with peer-reviewed research in robust learning, transfer/generative modeling, network analytics and wearable-sensor recognition. Previously taught undergraduate Computer Programming, Communication Systems and Wireless Communication; supported laboratory instruction and undergraduate/graduate research mentoring. Current industry ML engineer supporting AT&T service-assurance diagnostics, with earlier applied data science and cloud engineering work. Research and teaching interests span data science, machine learning, statistical computing, Python, software fundamentals, data engineering and trustworthy AI.

Research Areas

Trustworthy and Data-Efficient Machine Learning; Statistical Learning, Domain Shift and Model Reliability; Time-Series Data Science and Wearable Sensing; Anomaly Detection and Intelligent Network Analytics; Reproducible Data Science and Student-Centered Applied Research

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*

University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*

University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*

University of Engineering and Technology, Taxila, Pakistan

Doctoral Research

AI/ML for reliable wireless networks: transfer and generative propagation modeling, anomaly and outage detection, data-efficient learning, self-healing network optimization, and cross-domain robustness evaluation.

Teaching and Mentoring

University-Level Teaching

Lecturer, Department of Technology **12/2017 - 09/2019**

University of Lahore, Pakistan *Lahore, Pakistan*

- Taught programming, communication systems, and wireless communication with practical examples.
- Advised senior-design students on projects, academic plans, and early careers.
- Aligned course materials with learning outcomes and accreditation standards.
- Supported accreditation, exams, technical workshops, and student activities.

Teaching areas: Computer Programming and Computational Problem Solving (previous instructor experience); Introduction to Data Science and Python Data Analytics; Applied Machine Learning, Neural Networks and AI; Data Mining, Classification, Clustering and Statistical Modeling; Data Engineering and Cloud Analytics; Computer Networks and Communication Systems; Algorithms and Data Structures (with course-specific preparation); Capstone, Independent Research and Graduate Data Science Topics

Student mentoring and research development

- Mentored undergraduate and graduate projects involving machine-learning implementation, simulation, experimental design, statistical comparisons and technical presentations.
- Helped students establish reproducible baselines, scope realistic research questions and report model limitations alongside quantitative results.
- Supported hands-on learning with computational exercises, debugging, structured feedback and technical writing guidance.

Technical Skills, Engineering Methods & Research Tools

Machine Learning & Data Science: Python, NumPy, pandas, scikit-learn, PyTorch, TensorFlow, LightGBM, statistical inference, classification, regression, clustering, time series, transfer learning, domain adaptation, anomaly detection and reproducible evaluation

Programming & Computer Science: Python, C/C++, SQL, programming instruction, Git/GitHub, Linux, data structures and algorithms foundations, testing, data visualization and computational problem-solving

Cloud & Data Engineering: Azure Databricks, Apache Spark/PySpark, Azure Data Factory, Azure Synapse, Azure Functions, ETL/ELT, data quality, Azure DevOps CI/CD and cloud analytical pipelines

Research & Mentorship: Experimental design, simulations, published scholarship, technical writing, student-project supervision, cross-validation, open/synthetic datasets, research ethics and interdisciplinary collaboration

Domain Applications: Wireless and network data, service assurance, generative data modeling, sensor signals, healthcare time series and engineering reliability

Academic and Industry Research Experience

University Research Appointments

Graduate Research Assistant **09/2019 - 07/2025**
AI4Networks, University of Oklahoma *Tulsa, OK*

- Built 3GPP-compliant Python simulation tools and an AI-enabled 5G research testbed.
- Built GAN, transfer-learning, autoencoder, and reinforcement-learning models for wireless data.
- Led 100+ experiments testing wireless ML under domain shift and limited labeled data.
- Developed self-healing and outage-management methods, improving resilience by 30%.
- Published wireless AI research and mentored junior researchers in experimental design.

Research Associate, ComSens Lab **06/2016 - 12/2017**
COMSATS University Islamabad, Pakistan *Islamabad, Pakistan*

- Modeled wireless communication systems through simulation, analysis, and technical reports.
- Mentored Bachelor's and Master's theses in experiment design and research presentation.
- Reviewed student reports and manuscripts for rigor and publication readiness.
- Supported proposals, literature reviews, lab seminars, and research coordination.

Full-Time Research Scholar, Communication Systems Lab **08/2014 - 02/2016**
University of Engineering and Technology, Taxila, Pakistan *Taxila, Pakistan*

- Researched wireless communication systems while completing an MSc in Telecommunications.
- Supported communications courses, labs, assignments, grading, and examinations.
- Led wireless communication labs and prepared technical manuals and experiments.
- Developed course and student-project materials for practical wireless engineering.

Industrial Research and Engineering

Machine Learning Engineer **02/2025 - Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Applied domain-adaptive ML to telecom service assurance and model validation.
- Designed cross-domain evaluation and error-analysis methods for operational ML.
- Worked with engineering and public-safety teams on reliable network assurance.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**
Impac Exploration Services *Weatherford, OK*

- Built LLM-enabled analytics to interpret geochemical well data for decision support.
- Built Python pipelines in Azure Databricks, Functions, and Synapse Analytics.
- Automated exploration-data ETL, cutting preparation time by 40%.
- Used data-validation and quality controls to improve output reliability by 20%.
- Turned geologists' technical needs into structured datasets and useful analytics.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR logs with ML, reducing diagnostic time by 20%.
- Improved 5G NR-NTN cell selection by 15% through data-driven validation.
- Presented operational data insights to next-generation wireless engineering teams.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved wireless CSI compression by 25% with transfer learning and autoencoders.
- Ran 50+ experiments on wireless ML robustness under changing data conditions.
- Translated research results into recommendations for future wireless networks.

Advanced-Degrees Intern **06/2022 - 08/2022**
AT&T Labs *Bedminster, NJ*

- Improved 5G standalone user-timing precision by 20% with ML and statistics.
- Analyzed user traces with SQL and presented results to engineering leadership.
- Turned operational traces into indicators for network diagnosis and optimization.

Selected Peer-Reviewed Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches *HealthINF | Accepted* 2026

AI-Powered Resilience: A Dual-Approach for Outage Management in Dense Cellular Networks *Computer Communications* 2025 | DOI

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *Sensors 2025* 2025 | DOI

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *IEEE PIMRC* 2024 | DOI

Refining Wireless Propagation Models using Domain-Informed GANs amid Data Scarcity *IEEE VTC-Fall 2024*
2024 | Link

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *IEEE SmartNets*
2024

An AI-Enabled Framework to Defend Ingenious MDT-Based Attacks on the Emerging Zero Touch Cellular Networks *IEEE Network, 38(1), 228-237*
2024 | DOI

Positioning Error Impact Compensation through Data-Driven Optimization in User-Centric Networks *IEEE GLOBECOM 2023*
2023 | Link

Selected Research and Engineering Platforms

Wearable Intelligence for Health Monitoring *Research Project | 2024-2026*

Developed deep-learning workflows for recognizing medication-taking and dietary activities from smartwatch accelerometer and gyroscope signals in free-living settings. Evaluated windowing, feature extraction, CNN/LSTM-style sequence models, cross-user generalization, and recall-oriented performance for health-monitoring applications.

Tools & Methods: Wearable AI, Smartwatch IMU, Time Series, Deep Learning, CNN/LSTM, Feature Engineering, Cross-User Evaluation, Activity Recognition, Healthcare AI

Propagation Models & Digital Twins *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

Tools & Methods: Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020-2024*

Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.

Tools & Methods: Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

TurboRAN Experimental Platform *Research Infrastructure | 2020-2024*

Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled network conditions. Supported repeatable measurement collection, testbed experimentation, RAN optimization studies, and validation of emerging wireless-network algorithms.

Tools & Methods: 5G Testbeds, RAN, Network Measurement, Experiment Design, Wireless Prototyping, Multi-Band Systems, Performance Evaluation, RAN Optimization

AI for Telecom Network & Security Optimization *Certification Project | 2026*

Applied AI concepts to telecommunications network optimization, security analysis, and customer-experience improvement through a structured specialization project. Framed telecom use cases around operational data, AI-assisted analysis, optimization opportunities, and practical service-improvement recommendations.

Tools & Methods: AI for Telecommunications, Network Optimization, Telecom Security, Customer Experience, Python, TensorFlow, Operational Analytics, Service Improvement

Academic Service and Honors

Selected Academic Service

- Contributed to undergraduate teaching, assessment and instructional/laboratory materials.
- Supported collaborative research, student mentoring, technical seminars and peer-reviewed research dissemination.

Selected Honors

William H. Barkow Scholarship *2024*
University of Oklahoma, Electrical and Computer Engineering

Clyde L. Farrar Endowment Scholarship *2024*
University of Oklahoma

P.H. Robinson Fellowship *2023*
University of Oklahoma

FAS-Charles Forgan Scholarship *2023*
University of Oklahoma

Nettie Vincent Boggs Graduate Endowment *2022*
University of Oklahoma

SHS-Paul D. Massad Scholarship *2022*
University of Oklahoma

Cleo Cross Memorial Scholarship *2021*
University of Oklahoma

Marjorie Haskell Engineering Scholarship *2021*
University of Oklahoma

Fully Funded PhD / Graduate Research Assistantship *2019*
University of Oklahoma

Selected Professional Development

1. Machine Learning — Stanford/Coursera | Completed | Mar 2020
Supervised and unsupervised learning, regularization, model evaluation, and practical predictive modeling.

2. Python for Data Science, AI & Development — IBM/Coursera | Completed | Jul 2026
Python programming, data analysis, APIs, and reproducible data-science workflows.

3. Introduction to Deep Learning & Neural Networks with Keras — IBM/Coursera | Completed | Aug 2026
Neural networks, training workflows, optimization, loss functions, and model evaluation with Keras.

4. Generative AI Engineering and Fine-Tuning Transformers — IBM/Coursera | Completed | Aug 2026
Transformer architectures, model fine-tuning, embeddings, and generative-AI engineering.

5. Introduction to Computer Vision with TensorFlow — Google/Coursera | Completed | Aug 2024
Image processing and convolutional neural networks for computer vision using TensorFlow.